

Lab 12

2x2 Factorial ANOVA (Between-Groups)

Instructions

For this assignment you will be using data from a research study that looked at mothers who have been identified as being depressed. The researchers invited mothers and their youngest children to come to the lab and complete a 30-minute structured play activity.

During the session, the researchers collected the data found in the `parent_child_data.sav` dataset. Make sure you have the dataset open and review the variables before starting the analyses.

All three research hypotheses use between-groups factorial designs with the child study data, examining different combinations of independent variables and dependent variables.

Reminder: Please take screen shots of the results of your homework checker and include them at the end of this worksheet. Be sure they are all green (correct). Your TA will not grade the assignment without this.

Walkthrough (5 points)

RH1: BG Factorial Analysis - Walkthrough

None

RQ1: Does the effect of **family type** depend on **mother's treatment status** as they relate to the amount of **praise** given during the play session?

The dependent variable is `praise` (how many times the mother praises her child during the data collection period).

The IVs are `famtype` (Family Type: 2-parent family vs. mother-only family) and `tx` (Treatment Status of the Mother: no treatment vs. treatment received).

1. What are the three effects in this factorial design? Check the SPSS database -> Is each BG or WG?

List of effects:

2. Complete the factorial analysis and fill in the table below (don't forget the marginal means!)

Table 1

Family Type	No Treatment	Treatment	EMM: Family Type
2-Parent Family			
Mother-Only Family			

Note. EMM: Treatment Status

3. Find the results of the test of the interaction:

Find the results of the test of the interaction:

$F = \underline{\hspace{1cm}}$ $df = \underline{\hspace{1cm}}, \underline{\hspace{1cm}}$ $p = \underline{\hspace{1cm}}$ $MSe = \underline{\hspace{1cm}}$ Is there an interaction ???

Do we need an LSDmmd to compare cell means to describe the pattern of the interaction? $\underline{\hspace{1cm}}$ Why or why not?

If necessary, find the components for the LSDmmd computation:

conditions = $\underline{\hspace{1cm}}$ $n = \underline{\hspace{1cm}}$ $df \text{ error} = \underline{\hspace{1cm}}$ $MSe = \underline{\hspace{1cm}}$ $LSDmmd = \underline{\hspace{1cm}}$

Write up the interaction results below using the simple effects of Mother's Treatment Status for each Family Type – be sure to follow the example in the spss handout.

Write up:

Find the results for the test of the Main effect of Mother's Treatment Status:

Find the results for the test of the Main effect of Treatment Status

$F = \underline{\hspace{1cm}}$ $df = \underline{\hspace{1cm}}, \underline{\hspace{1cm}}$ $p = \underline{\hspace{1cm}}$ $MSe = \underline{\hspace{1cm}}$ Is there a main effect ???

Do we need an LSDmmd to compare marginal means to determine the pattern of the main effect? $\underline{\hspace{1cm}}$ Why or why not

So, is the main effect of Treatment Status descriptive or misleading?

Write up: Write up the main effects results below (including whether it is descriptive or misleading) – be sure to follow the example in the spss handout.

Write-up:

Find the results for the test of the Main effect of Family Type:

Find the results for the test of the Main effect of Family Type

$F = \underline{\hspace{1cm}}$ $df = \underline{\hspace{1cm}}, \underline{\hspace{1cm}}$ $p = \underline{\hspace{1cm}}$ $MSe = \underline{\hspace{1cm}}$ Is there a main effect ???

Do we need an LSDmmd to compare marginal means to determine the pattern of the main effect? $\underline{\hspace{1cm}}$ Why or why not

So, is the main effect of Family Type descriptive or misleading?

Your Turn (10 points)

RH2: BG Factorial Analysis - Your Turn #1

None

RQ2: Does the effect of **mother's age group** depend on **child's age group** as they relate to the amount of **direct commands** given during the play session?

The dependent variable is `direct` (how many direct commands the mother gives her child during the data collection period).

The IVs are `magegroup` (Mother's Age Group: 18-27 years old vs. 28-35 years old) and `cagegroup` (Child's Age Group: 2-3 years old vs. 4-5 years old).

1. What are the three effects in this factorial design? Check the SPSS database -> Is each BG or WG?

List of effects:

2. Complete the factorial analysis and fill in the table below (don't forget the marginal means!)

Table 2

Mother's Age Group	2-3 Years Old	4-5 Years Old	EMM: Mother's Age Group
18-27 Years Old			
28-35 Years Old			

Note. EMM: Child's Age Group

3. Find the results of the test of the interaction:

Find the results of the test of the interaction:

F = _____ df = _____, _____ p = _____ MSe = _____ Is there an interaction ???

Do we need an LSDmmd to compare cell means to describe the pattern of the interaction? _____ Why or why not?

If necessary, find the components for the LSDmmd computation:

conditions = _____ n = _____ df error = _____ MSe = _____ LSDmmd = _____

Write up the interaction results below using the simple effects of Child's Age Group for each Mother's Age Group – be sure to follow the example in the spss handout.

Write up:

Find the results for the test of the Main effect of Mother's Age Group:

Find the results for the test of the Main effect of Mother's Age Group

$F = \text{_____}$ $df = \text{____, ____}$ $p = \text{_____}$ $MSe = \text{_____}$ Is there a main effect ???

Do we need an LSDmmd to compare marginal means to determine the pattern of the main effect?

_____ Why or why not

So, is the main effect of Mother's Age Group descriptive or misleading?

Write up: Write up the main effects results below (including whether it is descriptive or misleading) – be sure to follow the example in the spss handout.

Write-up:

Find the results for the test of the Main effect of Child's Age Group:

Find the results for the test of the Main effect of Child's Age Group

$F = \text{_____}$ $df = \text{____, ____}$ $p = \text{_____}$ $MSe = \text{_____}$ Is there a main effect ???

Do we need an LSDmmd to compare marginal means to determine the pattern of the main effect?

_____ Why or why not

So, is the main effect of Child's Age Group descriptive or misleading?

RH3: BG Factorial Analysis - Your Turn #2

None

RQ3: Does the effect of **mother's treatment status** depend on **child's age group** as they relate to the amount of **negative feedback** given during the play session?

The dependent variable is *negat* (how many times the mother gives negative feedback to her child during the data collection period).

The IVs are *tx* (Treatment Status of the Mother: no treatment vs. treatment received) and *cagegroup* (Child's Age Group: 2-3 years old vs. 4-5 years old).

1. What are the three effects in this factorial design? Check the SPSS database -> Is each BG or WG?

List of effects:

2. Complete the factorial analysis and fill in the table below (don't forget the marginal means!)

Table 3

Treatment Status	2-3 Years Old	4-5 Years Old	EMM: Treatment Status
No Treatment			
Treatment			

Note. EMM: Child's Age Group

3. Find the results of the test of the interaction:

Find the results of the test of the interaction:

$F = \text{_____}$ $df = \text{____, ____}$ $p = \text{_____}$ $MSe = \text{_____}$ Is there an interaction ???

Do we need an LSDmmd to compare cell means to describe the pattern of the interaction? _____ Why or why not?

If necessary, find the components for the LSDmmd computation:

conditions = _____ $n = \text{_____}$ $df \text{ error} = \text{_____}$ $MSe = \text{_____}$ $LSDmmd = \text{_____}$

Write up the interaction results below using the simple effects of Child's Age Group for each Mother's Treatment Status – be sure to follow the example in the spss handout.

Write up:

Find the results for the test of the Main effect of Mother's Treatment Status:

Find the results for the test of the Main effect of Treatment Status

F = ____ df = ____, ____ p = ____ MSe = ____ Is there a main effect ???

Do we need an LSDmmd to compare marginal means to determine the pattern of the main effect?
____ Why or why not

So, is the main effect of Treatment Status descriptive or misleading?

Write up: Write up the main effects results below (including whether it is descriptive or misleading) – be sure to follow the example in the spss handout.

Write-up:

Find the results for the test of the Main effect of Child's Age Group:

Find the results for the test of the Main effect of Child's Age Group

F = ____ df = ____, ____ p = ____ MSe = ____ Is there a main effect ???

Do we need an LSDmmd to compare marginal means to determine the pattern of the main effect?
____ Why or why not

So, is the main effect of Child's Age Group descriptive or misleading?

Write up: Write up the main effects results below (including whether it is descriptive or misleading) – be sure to follow the example in the spss handout.

Write-up:

Grab screenshots of the homework checker screens, showing all of your input answers in green (correct). Paste them all here. Your TA will not grade the assignment without this.

Paste Homework Checker screenshots here: